

Lab 2 Cable-Actuated Finger

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I. OBJECTIVES AND EXPERIMENTAL SETUP

The purpose of this experiment was to evaluate how well the constant curvature model applies to the cable-actuated finger we fabricated. To begin, we created two molds from cardboard, assembling them with a printed SDM Finger template and a hot glue gun [1]. 1/16" tubing was placed between the finger segments to create channels for threading the cable, as shown in **Fig. 1**. We then mixed 25 milliliters of equal parts of Dragon Skin 10, Part A and Part B, and poured the mixture into the molds. The silicone was cured for 40 minutes at 70°C. Once cured, the fabricated finger, with the cable threaded through the open channels, is shown in **Fig. 2**. The fingers were secured in place using an acrylic jig, with the cable from each actuator attached to a servo motor. We connected the servo motor to an Arduino and uploaded the control code from Blackboard. The Arduino code was modified to bend the finger at five different angles: 60, 90, 120, 150, and 180 degrees. Video footage was captured for each angle, with graph paper in the background as a reference frame. The final testing setup is shown in **Fig. 3**. To validate whether the finger followed the constant curvature model, finger positions—end effector, joint 1, and joint 2—labeled in **Fig. 4** were plotted in MATLAB.



Figure 1: Finger mold used to fabricate the cable-actuated finger.



Figure 2: Fabricated finger used for cable actuation.

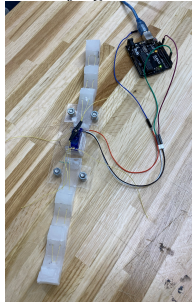


Figure 3: Experimental testing setup for cable actuating fingers utilizing Arduino and servo motor.

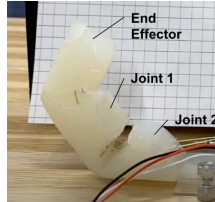


Figure 4: Fabricated finger with labels (end effector, joint 1, joint 2)

II. MATH

The cable-actuated soft finger operates under the assumption of constant curvature during actuation. This simplifies the analysis and allows for more manageable modeling of its behavior. The curvature of the finger is a central component, calculated using (1).

$$K = \frac{l_c - l_2}{l_c d} \quad (1)$$

where K is the curvature, l_c is the average length of the cable, l_2 is the remaining length of the cable after pulling, and d is the distance from the cable to the bottom of the finger.

$$\theta = K l_c \quad (2)$$

The bending angle of the finger θ is determined by (2). This relationship allows us to calculate the total bending of the finger based on the amount of cable pulled.

$$\theta_i = \frac{i}{n} \theta \quad (3)$$

Moreover, the finger is modeled as a series of joints, and the angle at each joint is a fraction of the total bending angle, divided equally across the number of segments. For each joint, the angle θ_i is calculated by (3), where n is the number of segments, and i is the index of the segment. This model follows the piecewise constant curvature assumption, which approximates the entire finger as a series of small, mutually tangent arcs, as discussed in the continuum robot modeling framework.

$$T = \begin{bmatrix} \cos\phi \cos\theta & -\sin\phi & \cos\phi \sin\theta & r \cos\phi (1 - \cos\theta) \\ \sin\phi \cos\theta & \cos\phi & \sin\phi \sin\theta & r \sin\phi (1 - \cos\theta) \\ -\sin\theta & 0 & \cos\theta & r \sin\theta \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

To model the finger's tip pose (position and orientation), we can use a transformation matrix, T displayed in (4) parameterized by arc parameters such as backbone curvature, κ out-of-plane rotation, ϕ and arc length l . Using (4), we can determine the position and orientation of the tip based on the bending behavior of the soft finger [2].

TABLE I. END EFFECTOR POSITIONS

Degrees of Actuation	Experimental X-Position [cm]	Experimental Y-Position [cm]	CCM X-Position [cm]	CCM Y-Position [cm]	CCM (Adjusted) X-Position [cm]	CCM (Adjusted) Y-Position [cm]	CCM (Adjusted and X-Corrected) X-Position [cm]	CCM (Adjusted and X-Corrected) Y-Position [cm]
60	6.1	0.94	-0.72	6.56	11.83	1.4	6.83	1.4
90	6.06	1.86	-2.04	1.49	11.67	2.07	6.67	2.07
120	5.76	2.72	0.62	0.13	11.45	2.74	6.45	2.74
150	4.48	4.12	1.02	1.78	11.19	3.38	6.19	3.38
180	2.74	4.68	-0.55	1.72	10.87	3.99	5.87	3.99

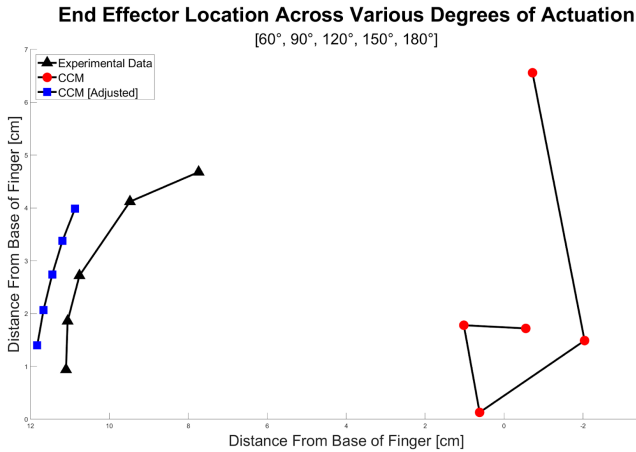


Figure 5: Plot of finger end effector positions for experimental, constant curvature model and constant curvature model with correction factor for five initial angles of actuation.

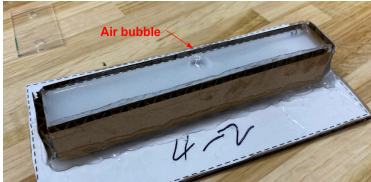


Figure 6: Air bubbles in mold after cure.

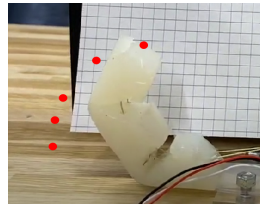


Figure 7: Plot of end effector position over 5 levels of actuation.

III. RESULTS AND DISCUSSION

The purpose of this experiment was to assess the validity of the constant curvature assumption in a cable-actuated soft finger. **Table 1** displays the experimental, corrected, and constant curvature model end effector positions obtained for various angles. As seen in **Fig. 5**, the fingers did not consistently follow the constant curvature assumption when actuated. Several factors, such as imperfections in the fabrication process (air bubbles—**Fig. 6**—, inconsistencies in tension), contributed to deviations from the model. As observed in **Fig. 7**, the higher the initial tension in the cable, the more closely the finger followed the constant curvature model. With lower tension, the finger tended to curl less, leading to worse performance. When comparing experimental data to the constant curvature model (**Fig. 5**), the model did not accurately predict the fingertip's position, particularly at lower initial angles. While applying a constant correction factor could help adjust the data, we observed that the correction factor varied depending on the starting angle (0.06 for 60 degrees, 0.07 for 120 degrees, and 0.25 for 180 degrees). The average correction factor was 0.07, but variations are likely due to fabrication inconsistencies (e.g., air bubbles, cable tightness). If a small weight is attached to the tip of the finger, its alignment with the constant curvature model will worsen, as the weight creates uneven force distribution along the finger. Similarly, if a force is applied to the side of the finger, it will likely buckle at unintended points, deviating from the expected behavior predicted by the constant curvature model. If the cable were positioned on the opposite side of the actuator, the result would likely deviate from the constant curvature model. In this orientation, activating the actuator would only loosen the cable, resulting in no tension being applied to pull the finger upwards. Without this tension, the finger would remain stationary, and the constant curvature model, which relies on tension to create the expected bend, would not be an accurate fit. In both cases, the performance will not match the model's assumptions.

REFERENCES

- [1] Soft Robotics Toolkit, "SDM Design," [Online]. Available: <https://softroboticstoolkit.com/book/sdm-design>.
- [2] J. Rogatinsky, "Constant Curvature Modeling Approach," presented at the Morphable Biorobotics Lab, Feb. 14, 2024.